



MT1008 – Multivariable Calculus

Assignment No: 01

CLO: 01

Section: BCS

Semester: Spring-26

Date: 12-02-2026 before 02:00pm

Marks: 120

Instructions:

1. Plagiarized work will result in zero marks.
 2. No retake or late submission will be accepted.
 3. Attach complete code, results, and screenshot for questions that require programming solutions. Programs/codes should be typed.
 4. **Assignment is to be submitted in soft copy as well as in hardcopy. Submit the hardcopy before the deadline through CR, and softcopy on GCR.**
 5. The softcopy should be a single PDF file of your assignment including programming and non-programming questions.
 6. The PDF file should be according to the following format: id_section_A1 e.g., i25-123456_A_A1. A1 in the end denotes Assignment 1. **The title page must include complete student information, including name, section, id, course name, and assignment number.**
 7. The images of by-hand solution should be properly scanned. You can use any mobile application such as CamScanner or Adobe Scan for scanning. Each of these applications allows you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
 8. Python is the only approved programming language.
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Question 1

In machine learning, we often minimize a Loss Function to train a model. Consider a simplified loss surface defined by the function:

$$f(x, y) = \ln(1 + e^{-(ax+by)})$$

where x, y are model weights, and $a = 2, b = -3$ are fixed constants.

- i. Calculate the first-order partial derivatives.
- ii. Express the gradient $\nabla f(x, y)$ at the specific point $P(1,1)$.
- iii. Find the rate of change of f at point $P(1,1)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.
- iv. Write a Python script using *NumPy*, *Matplotlib*, and *SymPy* to perform the following:
 - a. Use *SymPy* to compute the partial derivatives and compare them with your manual results.
 - b. Generate a contour plot of $f(x, y)$ over the domain $x, y \in [-2, 2]$. Overlay the gradient vector field using *plt.quiver*.
- v. Based on your gradient calculation at $(1, 1)$, which variable (x or y) has a greater impact on reducing the loss? Explain this in the context of the magnitudes of a and b .

Question 2

A heat map represents the temperature distribution $T(x, y)$ on a CPU die during a heavy computational load. The temperature is modeled by the function:

$$T(x, y) = 40 + 60e^{-(x^2+2y^2)}$$

where T is in Celsius and x, y are coordinates on the chip.

- i. Find the general equation for the level curves $T(x, y) = k$. What geometric shapes do these curves represent?
- ii. Sketch and label the level curve (isotherm) for $T = 70^\circ\text{C}$. Solve for y in terms of x .
- iii. Calculate the gradient ∇T at point $(1, 0.5)$. Show analytically that this gradient vector is perpendicular to the tangent of the level curve passing through that point.
- iv. Use *plt.contour()* to plot level curves for $k = [45, 55, 65, 75, 85, 95]$.
- v. Using the plot, identify the areas of the chip with the highest "thermal gradient" (where the level curves are most packed). How does the coefficient of y^2 in the exponent affect the shape of the cooling zones compared to x^2 ?

Question 3

In computer graphics and game development, Implicit Surfaces are used to define 3D objects. Consider the potential field V generated by two data clusters in a 3D feature space:

$$V(x, y, z) = x^2 + y^2 - z^2$$

- i. Describe the level surface $V(x, y, z) = c$ for:
 - a. $c > 0$ (e. g., $c = 1$)



- b. $c = 0$
- c. $c < 0$ (e.g., $c = -1$)
- ii. Find the equation of the tangent plane to the level surface $V = 4$ at the point $(2, 1, 1)$.
- iii. Use the *skimage.measure.marching_cubes* algorithm or *Plotly* to render the level surface $V = 1$ in a 3D coordinate system.

Question 4

In image processing, we often use filters that average pixel values. Suppose this function represents the weight distribution of a filter near a central pixel at $(0,0)$. For the filter to be "stable," the limit must exist and be equal to the function's value at that point.

$$f(x, y) = \begin{cases} \frac{10xy^2}{x^2 + y^2}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$$

- i. Test the limit as $(x, y) \rightarrow (0,0)$ by taking various paths.
- ii. Since path testing doesn't prove a limit exists, use polar coordinates and squeeze theorem to conclude that the limit exists.
- iii. Use *plot_surface* to visualize the function.
- iv. Explain why L'Hôpital's rule cannot be directly applied to multivariable limits and why the Squeeze Theorem is the go-to tool while dealing with convergence in algorithms.

Question 5

Suppose a signal z in a computational network is determined by two intermediate parameters, u and v . The relationship is given by:

$$z = f(u, v) = u^2 e^v$$

However, u and v are not independent; they evolve over time ' t ' according to:

$$u(t) = \sin(2t) \text{ and } v(t) = \cos(t)$$

- i. Use the Chain Rule to find an expression for the rate of change of the signal with respect to time.
- ii. Calculate the exact value of $\frac{dz}{dt}$ at the instant $t = \frac{\pi}{4}$.
- iii. Calculate the values of u and v at $t = \frac{\pi}{4}$, and then calculate z . Does z appear to be increasing or decreasing at this moment?
- iv. Use *sympy.diff* to compute the derivative of z with respect to t directly by substituting $u(t)$ and $v(t)$ into the original function. Compare this direct substitution result with the chain rule result.
- v. Plot $z(t)$ over the interval $t \in [0, \pi]$ and mark the point where $t = \frac{\pi}{4}$.
- vi. At $t = \frac{\pi}{4}$, which intermediate variable (u or v) is contributing more to the change in z ?



- vii. In programming, if z depended on 10,000 variables instead of just two, why is Chain Rule more computationally efficient than Direct Substitution?

Question 6

A tech firm manufactures two models of a high-performance router: The alpha-X and the beta-Y. The market price of these routers depends on their scarcity. The price P (in hundreds of dollars) for the alpha-X is given by $P = 50 - 2x$, where x is the quantity sold (in thousands). The price Q (in hundreds of dollars) for the beta-Y is given by $Q = 80 - 4y$, where y is the quantity sold (in thousands). The joint manufacturing cost $C(x, y)$ accounts for shared resources and overhead:

$$C(x, y) = 2x^2 + 2xy + y^2 + 100$$

Determine the revenue function $R(x, y)$ and the profit function $P(x, y)$.

Question 7

You are designing cloud architecture with three existing data centers located at coordinates $A(0,0)$, $B(1,0)$, & $C(0,1)$. A new central hub is to be placed at an unknown location $D(x, y)$ to manage traffic between them. The total system latency $L(x, y)$ is determined by the following constraints:

- Traffic weights: the latency contribution from each data center is proportional to the square of its distance from the hub (d^2), multiplied by a load factor based on the traffic volume:
 - Data center A (low traffic): Load factor = 1
 - Data center B (medium traffic): Load factor = 2
 - Data center C (high traffic): Load factor = 3
- Overhead: There is a fixed processing delay (constant overhead) of 5 units regardless of location.

Construct a simplified multivariable function $L(x, y)$ representing the total system latency. Show your work by setting up weighted sum and expanding the algebraic terms to the form $Ax^2 + By^2 + Cx + Dy + E$.

Question 8

You are optimizing the design of a fiber data duct with a trapezoid cross-section. The data transmission bandwidth is proportional to the duct's cross-sectional area. You have a fixed length L of shielding material to cover the bottom and two slanted sides. Construct the multivariable function $A(x, \theta)$ for the cross-sectional area, where x is the length of the slanted sides and θ is the angle from the horizontal.



Question 9

In machine learning, a feature transformation is defined as:

$$\phi(x, y) = \left(\ln(1 - x^2 - y^2), \sqrt{y - x}, \frac{1}{x + y - 2} \right)$$

- i. Find the domain of function ϕ .
- ii. Check if region is open, closed, both, or neither.
- iii. Check whether it is bounded or unbounded.
- iv. Sketch the region.

Question 10

A robot's sensor is modeled by:

$$S(x, y) = \sqrt{4 - (x - 1)^2 - (y + 2)^2}$$

- i. Find the domain of function S .
- ii. Check if region is open, closed, both, or neither.
- iii. Check whether it is bounded or unbounded.
- iv. Sketch the region.

Question 11

Consider the Cobb-Douglas production function:

$$P(L, K) = 1.5L^{0.6}K^{0.4}$$

where L = labor hours, K = capital investment.

Using Python:

- i. Determine and sketch domain ($L > 0, K > 0$).
- ii. Compute marginal productivity of labor $\left(\frac{\partial P}{\partial L}\right)$ and capital $\left(\frac{\partial P}{\partial K}\right)$.
- iii. Verify symmetry of mixed partial derivatives.
- iv. Plot production surface for $L \in [1, 100], K \in [1, 100]$.

Question 12

Using Python, compute all first and second partial derivatives (including mixed partials) for the following functions:

- i. $z = 10 - x^2 + y^2 + 3 \cos(3(x^2 + y^2))$
- ii. $g(x, y, z) = yx^3 + xyz - e^{y^2z}$
- iii. $h(x, y) = \cos(\sin x + y)$